

PRAKHAR DWIVEDI

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CAREER OBJECTIVE

Third-year undergraduate with hands-on experience in ROS 2-based robotics development through competitive rover challenges. Passionate about autonomous systems, robot modeling, and sensor integration. Eager to deepen my understanding of robotics software stacks and contribute to impactful engineering projects.

EDUCATION

Sardar Vallabhbhai National Institute of Technology, Surat

B.Tech in Electronics and VLSI Engineering

Senior Secondary (ISC)

Percentage: 94.75%

Expected 2028

CGPA: 8.56/10.0

2023

TECHNICAL SKILLS

- **Programming:** C, Python, Java, C++, ARM Cortex-M Assembly (Basic)
- **Robotics Frameworks:** ROS 2, Gazebo, RViz, URDF/Xacro modeling, Nav2, MoveIt2, Slam Toolbox
- **Sensors & Perception:** LiDAR, IMU, Camera integration, sensor fusion pipelines
- **Embedded Systems:** Arduino, ESP32, STM32
- **Tools:** Linux, Git/GitHub, VS Code, STM32CubeIDE
- **Protocols:** I2C, SPI, UART, CAN

EXPERIENCE

ROS 2 Developer – Team Drishti | International Rover Challenge (IRC) 2026

Ranked 9th Globally

Workspace: github.com/jivalz/NOIR-26

Moevit Configured Custom Robotic Arm : github.com/prakhar-29/Robotic_arm_Noir

- Integrated ROS 2 (Humble) middleware stack for autonomous rover operations, enabling communication between navigation, perception, and manipulation subsystems
- Built a URDF/Xacro model of the team's 6-DOF robotic arm from CAD data, defining link geometries and joint structure; visualized and tested the model in RViz using joint state publishers
- Gained hands-on understanding of how robot description files translate physical arm structure into a ROS 2-compatible simulation model
- Developed sensor integration pipelines for LiDAR, cameras, and IMU to support obstacle detection and localization in outdoor competition environments
- Worked on autonomous navigation logic and path planning for competition tasks including terrain traversal and sample collection
- Debugged multi-threaded ROS 2 nodes and sensor data pipelines to improve system stability during field testing
- Collaborated with teammates on object detection, autonomy algorithms, and visualization tooling for rover telemetry and mission monitoring

RELEVANT COURSEWORK

- Digital Logic Design • Analog Circuits • Control Systems • Microprocessors & Microcontrollers • Digital Signal Processing
- Data Structures & Algorithms

ACADEMIC ACHIEVEMENTS

- **JEE Main 2024:** 98.9 Percentile
- **JEE Advanced 2024:** All India Rank 13,580

EXTRACURRICULAR

- **Silver Medal:** Under-17 Cricket Tournament (UP & UK), 2018 – CISCE
- **Silver Medal:** Under-17 Zonal Cricket Tournament, 2018 – CISCE